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Model Complexity, Expectations, and Asset Prices

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1 Big picture

- **Question.** How do limits on the complexity of agents' statistical models affect expectations, asset prices, return predictability, and forecast-error predictability?
- **Core friction.** The true data-generating process may require n hidden factors, but agents can only entertain and update beliefs over models with at most k factors.
- **Behavioural assumption.** Agents are Bayesian and observe the realized fundamental sequence, but their prior has support only on models with complexity no higher than k .
- **Key benchmark.** If $k \geq n$, the true model is in the agents' model class, and long-run beliefs coincide with rational expectations.
- **Main departure.** If $k < n$, the model class is misspecified. Long-run Bayesian learning selects the k -factor approximation with minimum Kullback–Leibler divergence from the truth.
- **Asset pricing link.** Prices satisfy a no-arbitrage-style equation $y_t = x_t + \delta E_t[y_{t+1}]$. Excess returns are discounted forecast revisions.
- **Main theoretical object.** The paper characterizes the whole term structure of return predictability and forecast-error predictability as a function of the true autocorrelation function and the complexity constraint k .
- **Important implication.** Complexity constraints can generate predictable excess returns and predictable forecast errors even though agents are Bayesian and internally consistent.
- **Single-factor intuition.** With $k = 1$, agents fit a simple exponentially decaying autocorrelation to a richer process. If the true autocorrelation is hump-shaped or oscillatory, return predictability can reverse sign across horizons.
- **Empirical application.** The framework is applied to uncovered interest rate parity (UIP). The fundamental is the interest rate differential, and the asset return is the currency excess return.
- **Main empirical message.** A one-factor subjective model can simultaneously match the short-run forward discount puzzle and the long-run predictability reversal puzzle.
- **Forecast survey evidence.** Survey-based forecast errors of interest-rate differentials are also predictable in the pattern implied by a one-factor model.
- **Takeaway.** A small model-complexity constraint can generate asset-pricing patterns that look like belief distortions, but the distortions are endogenous consequences of Bayesian learning under an under-dimensional model class.

2 Data and measurement

2.1 Objects in the theory

- x_t is the exogenous fundamental. In the general framework it can be an asset dividend, a macro state, or the interest rate differential in the UIP application.
- y_t is the endogenous price, pinned down by the subjective-expectation pricing equation.
- rx_{t+1} is the one-period excess return.
- n is the number of hidden factors needed to represent the true process.
- k is the maximum number of hidden factors agents can entertain.

Interpretation: The paper treats complexity as model dimension. A higher k means agents can store and update a richer set of state variables.

2.2 UIP application data

- The UIP application follows Engel-style horizon regressions using monthly exchange-rate and interest-rate data.
- The return-predictability sample runs from January 1985 to December 2021.
- The currency return is a trade-weighted average against the US dollar.
- Countries in the return-predictability application include Australia, Canada, the Eurozone/Germany before the euro, New Zealand, Japan, Switzerland, the United Kingdom, and the United States.
- Interest-rate differentials are computed from covered-interest-parity implied forward rates and monthly exchange-rate data.
- The dependent variable is the h -month currency excess return.
- The predictor is the current interest-rate differential.

Interpretation: A positive short-horizon coefficient means high interest-rate currencies earn positive excess returns; a negative long-horizon coefficient means the UIP violation reverses.

2.3 Survey expectations data

- Forecast-error predictability is studied using Consensus Economics survey forecasts.
- The survey sample runs from June 1998 to December 2021.
- Survey respondents forecast country-specific interest rates.
- The authors form consensus forecasts using the median response for each country.
- They focus on three-month-ahead forecasts and compare survey forecast errors with model-implied forecast errors.
- Australia and New Zealand are dropped from the survey exercise because the authors do not have forecasts for them.

Interpretation: Survey data make the expectations channel directly testable: the model is not only about returns, but also about forecast errors.

2.4 No empirical tables

The paper has no main empirical tables. Its visual evidence consists of four main figures and one appendix figure. The most important “tables” for studying the paper are its equation, proposition, and theorem panels, which are extracted and grouped in the notes below.

3 Models and empirical specifications

3.1 True n -factor data-generating process

The true fundamental is generated by

$$z_t = A^* z_{t-1} + B^* \varepsilon_t, \quad x_t = (c^*)' z_t. \quad (1)$$

The eigenvalues of A^* lie inside the unit circle, so the process is stationary.

Interpretation: The true process can have a complicated low-frequency autocorrelation pattern if it needs many hidden factors or has complex eigenvalues.

3.2 Pricing equation and excess returns

Prices satisfy

$$y_t = x_t + \delta E_t[y_{t+1}], \quad (2)$$

and excess returns are

$$rx_{t+1} = \delta y_{t+1} - y_t + x_t. \quad (3)$$

Using the pricing equation,

$$rx_{t+1} = \delta(y_{t+1} - E_t[y_{t+1}]).$$

Interpretation: Excess returns are forecast revisions. If agents misperceive the intertemporal structure of x_t , those forecast revisions can be predictable.

3.3 Agents' constrained k -factor model

Agents can only consider subjective models of the form

$$\omega_t = A\omega_{t-1} + B\varepsilon_t, \quad x_t = c'\omega_t, \quad (4)$$

where $\omega_t \in \mathbb{R}^k$.

Interpretation: The restriction is only on dimension. Conditional on k , all model parameters are determined by Bayesian learning from the data.

3.4 Bayesian learning under misspecification

Agents start with a prior over k -factor models. With a long history of observations, posterior beliefs concentrate on models that minimize KL divergence from the true process:

$$KL(\theta^*||\theta) = E^*[-\log f^\theta(x_{t+1}|x_t, \dots)] - E^*[-\log f^*(x_{t+1}|x_t, \dots)].$$

Interpretation: When $k < n$, agents do not learn the true model; they learn the best feasible low-dimensional approximation.

3.5 Return-predictability regression

The horizon- h return-predictability regression is

$$rx_{t+h} = \alpha_h^{rx} + \beta_h^{rx} x_t + \varepsilon_{t,h}. \quad (5)$$

Interpretation: The sequence $\{\beta_h^{rx}\}_{h \geq 1}$ is the term structure of return predictability. Under rational expectations, all these coefficients are zero.

3.6 Forecast-error predictability regression

The forecast-error regression is

$$x_{t+h+m-1} - E_{t+h-1}[x_{t+h+m-1}] = \alpha_{h,m}^{fe} + \beta_{h,m}^{fe} x_t + \varepsilon_{t,h,m}. \quad (6)$$

Interpretation: This asks whether today's fundamental predicts future subjective forecast errors. Under rational expectations, all $\beta_{h,m}^{fe}$ are zero.

3.7 Characterization using true and subjective autocorrelations

Let ξ_h^* denote the true autocorrelation of x_t at lag h and ξ_h denote the autocorrelation perceived by agents. The first key result expresses β_h^{rx} and $\beta_{h,m}^{fe}$ using the wedge between ξ_h^* and ξ_h .

Interpretation: The paper’s mechanism is not arbitrary optimism or pessimism. It is autocorrelation misspecification caused by a complexity constraint.

3.8 Theorem 1: complexity maps into predictability

Theorem 1 combines Bayesian learning under misspecification with the predictability formulas. It says the subjective autocorrelation function is generated by the best k -factor approximation, and the resulting return and forecast-error coefficients are functions of that approximation.

Interpretation: This is the main bridge between the behavioural constraint k and the observable asset-pricing regressions.

3.9 Rational expectations as a special case

If agents can contemplate models at least as rich as the true data-generating process, $k \geq n$, then the true model is in their model class:

$$\beta_h^{rx} = 0, \quad \beta_{h,m}^{fe} = 0 \quad \text{for all } h, m.$$

Interpretation: Predictability arises only when model complexity is binding.

3.10 Single-factor closed form

For $k = 1$, the return and forecast-error slopes have simple forms:

$$\begin{aligned} \beta_h^{rx} &= \frac{\delta}{1 - \delta \xi_1^*} (\xi_h^* - \xi_{h-1}^* \xi_1^*), \\ \beta_{h,m}^{fe} &= \xi_{h+m-1}^* - (\xi_1^*)^m \xi_{h-1}^*. \end{aligned}$$

Interpretation: The one-factor model compares the true h -period autocorrelation with the autocorrelation implied by a simple exponential process. This is why hump-shaped or oscillatory ACFs can produce sign reversals.

3.11 Identifying complexity from forecasts

If subjective forecasts can be written as

$$E_t[x_{t+m}] = \sum_{\tau=1}^{\infty} \phi_{\tau}^{(m)} x_{t+1-\tau},$$

then the Hankel matrix with elements $\Phi_{ij} = \phi_{i+j-1}^{(m)}$ has rank

$$\text{rank}(\Phi) = k.$$

Interpretation: Forecast data can be used to infer the dimension of agents’ subjective model. In the UIP survey application, the evidence is consistent with $k = 1$.

4 Identification and empirical design

4.1 What identifies the theoretical mechanism

- The theory compares objective and subjective autocorrelation functions.
- The subjective autocorrelation is not freely estimated; it is pinned down by the best k -factor approximation.
- Return predictability follows from the wedge between the true and subjective autocorrelation functions.
- Forecast-error predictability provides a separate test of the belief mechanism.

Interpretation: The key empirical discipline is that the same estimated autocorrelation function of interest-rate differentials must explain both return predictability and forecast-error predictability.

4.2 Why UIP is a useful application

- UIP has two robust puzzles: high interest-rate currencies earn positive excess returns at short horizons, but long-horizon predictability reverses sign.
- These two facts are hard to reconcile with simple constant risk-premium explanations.
- The paper's model can generate both from a single source: low-dimensional subjective forecasting.
- The empirical return regressions and the model-implied coefficients are not fitted with a free behavioural bias parameter; the coefficients are determined by k and the empirical ACF.

Interpretation: UIP is a demanding test because the model must match the full term structure of predictability, not only the short-run Fama coefficient.

4.3 Why survey evidence matters

- Return predictability alone may reflect risk premia.
- Forecast-error predictability is directly about expectations.
- The survey evidence asks whether interest-rate differentials predict future forecast errors in the same pattern as the model.
- The one-factor model matches the sign pattern and approximate shape of survey-based forecast-error predictability.

Interpretation: The survey results strengthen the interpretation that the return pattern is connected to expectations, not only to unobserved risk premia.

4.4 What the paper cannot fully prove

- The application is not a randomized empirical design.
- The framework is reduced form and abstracts from time-varying fundamental risk premia.
- The parameter k is treated as an exogenous behavioural primitive rather than an endogenous choice.
- The main analysis focuses on a representative-agent environment, with heterogeneity handled in an appendix extension.
- The model studies one time series at a time, so cross-sectional asset-pricing implications are left for future work.

Interpretation: The paper's claim is not that risk premia are absent. Its claim is that model-complexity constraints can generate the observed expectation and return patterns in a disciplined way.

5 Figures and extracted equation/proposition panels

The original paper contains figures but no empirical tables. To keep the notes compact, I group adjacent figures and key proposition/equation panels in pairs or triples.

5.1 Core environment: true model, prices, returns, and subjective model

2.1. Reduced-form framework

Consider a discrete-time economy consisting of a unit mass of identical agents. Agents form subjective expectations about an exogenous sequence of variables, $\{x_t\}_{t=-\infty}^{\infty}$, which we refer to as the economy's *fundamentals*. Depending on the context, the fundamental may correspond to an asset's dividend stream over time, interest rate differential between two countries, or any sequence of variables that can be treated as exogenous in that specific application.

The sequence of fundamentals $\{x_t\}_{t=-\infty}^{\infty}$ is generated by a stationary n -factor model given by

$$z_t = \mathbf{A}^* z_{t-1} + \mathbf{B}^* \varepsilon_t$$

(a) True n -factor process

While at any given time t , agents observe the sequence of realized fundamentals up to that time, we assume that they observe neither the underlying factors $(z_1, \dots, z_n)$ nor the collection of parameters $\theta^* = (\mathbf{A}^*, \mathbf{B}^*, c^*)$ that governs the data-generating process. As a result, they use the past realizations of the fundamental to learn about the underlying process that generates $\{x_t\}_{t=-\infty}^{\infty}$, which they then use to make forecasts about the future.

As our main behavioural assumption, we assume that agents face a constraint on the complexity of statistical models they can consider. Specifically, as in Molavi (2022), we assume that they can only entertain models with at most k factors, where k may be distinct from the true number of factors, n , in equation (1). The number of factors, k , thus indexes the agents' degree of sophistication, with a larger k corresponding to agents who can entertain a richer class of models.

Formally, we assume that agents can only hold and update beliefs over the set of k -factor models $\Theta_k = \{(\mathbf{A}, \mathbf{B}, c) : \mathbf{A}, \mathbf{B} \in \mathbb{R}^{k \times k}, \rho(\mathbf{A}) < 1, \text{ and } c \in \mathbb{R}^k\}$ of the following form:

(a) Agents' constrained k -factor model

$$y_t = x_t + \delta \mathbb{E}_t[y_{t+1}], \quad (2)$$

where $\mathbb{E}_t[\cdot]$ denotes the agents' time t subjective expectation and $\delta \in (0, 1]$ is a constant. Equation (2) is akin to a standard no-arbitrage condition with a natural interpretation. For example, if the sequence $\{x_t\}_{t=-\infty}^{\infty}$ represents an asset's dividend stream, then y_t corresponds to the price of the asset at time t , with equation (2) simply capturing the fact that the asset's price is the sum of its dividends at that time and its expected future price, discounted at some rate δ . We also note that agents' subjective expectations, $\mathbb{E}_t[\cdot]$, may be distinct from the expectations arising from the true data-generating process (1), which we denote by $\mathbb{E}_t^*[\cdot]$.

Given the sequence of fundamentals and prices $\{(x_t, y_t)\}_{t=-\infty}^{\infty}$, we define *excess returns* at time $t+1$ as the sum of the fundamental and the change in price between t and $t+1$, properly discounted:

(b) Price and return equations

$$rx_{t+1} = \sum_{\tau=1}^{\infty} \delta^\tau (\mathbb{E}_{t+1}[x_{t+\tau}] - \mathbb{E}_t[x_{t+\tau}]). \quad (5)$$

For the rest of the paper, we restrict our attention to agents' long-run beliefs by assum-

(b) Returns as forecast revisions

Read:

- The true process has n hidden factors, summarized by $(\mathbf{A}^*, \mathbf{B}^*, c^*)$.
- Prices depend on subjective expectations, not necessarily true expectations.
- Excess returns are realized forecast revisions.
- Agents' subjective models have only k hidden factors.

Economic message: The source of predictability is not irrational updating. It is that prices depend on forecasts from a lower-dimensional model.

5.2 Predictability regressions and Proposition 1

quantify the extent to which the current realization of the fundamental predicts future excess returns at different horizons:

$$rx_{t+h} = \alpha_h^{rx} + \beta_h^{rx} x_t + \varepsilon_{t,h}, \quad (6)$$

where $h \geq 1$ denotes the horizon and rx is the excess return, as defined in equation (3). If returns are unpredictable—as would be the case under rational expectations—then $\beta_h^{rx} = 0$ for all horizons h . If, on the other hand, an increase in the fundamental today is associated with higher excess returns at horizon h , then $\beta_h^{rx} > 0$. Thus, the family of slope coefficients $(\beta_1^{rx}, \beta_2^{rx}, \dots)$ not only provides a measure for departures from the rational expectations benchmark but also determines the extent to which such departures vary with horizon h .⁹

3.2. Forecast-error predictability

We test for the predictability of forecast errors by relying on a similar family of linear regressions. These regressions, which resemble the specification in Kothari and Walther (2021), measure the extent to which the current realization of the fundamental predicts m -step-ahead forecast errors at a given horizon $h \geq 1$:

$$x_{t+h+m-1} - \mathbb{E}_{t+h-1}[x_{t+h+m-1}] = \alpha_{h,m}^{fe} + \beta_{h,m}^{fe} x_t + \varepsilon_{t,h,m}, \quad (7)$$

(a) Return and forecast-error regressions

which may be distinct from ε_t . The following result expresses the large-sample limits of the slope coefficients of regressions (6) and (7) in terms of these autocorrelations.

Proposition 1. *The slope coefficients of the return and forecast-error predictability regressions (6) and (7) are given by*

$$\beta_h^{rx} = \delta \frac{(\zeta_h^* - \zeta_h) - \sum_{\tau=1}^{\infty} \sum_{s=1}^{\infty} \Xi_{\tau s}^{-1} \zeta_s^* (\zeta_{h-\tau}^* - \zeta_{h-\tau})}{1 - \sum_{\tau,s=1}^{\infty} \delta^\tau \Xi_{\tau s}^{-1} \zeta_s^*} \quad (8)$$

and

$$\beta_{h,m}^{fe} = (\zeta_{h+m-1}^* - \zeta_{h+m-1}) - \sum_{\tau=1}^{\infty} \sum_{s=1}^{\infty} \zeta_{s+m-1}^* \Xi_{\tau s}^{-1} (\zeta_{h-\tau}^* - \zeta_{h-\tau}), \quad (9)$$

respectively, where Ξ is an infinite-dimensional matrix such that $\Xi_{ij} = \zeta_{i-j}^*$ for all $i, j \geq 1$.

This result has a few immediate implications. First, it is straightforward to verify that, if $\mathbb{E}[\cdot] = \mathbb{E}^*[\cdot]$, then $\zeta_\tau^* = \zeta_\tau^*$ for all τ and hence, $\beta_h^{rx} = \beta_{h,m}^{fe} = 0$ at all horizons $h \geq 1$. Thus, when agents use the correct model to make forecasts about the future realizations of the fundamental—as would be the case with rational expectations—neither excess returns nor forecast

(b) Proposition 1: coefficients from ACF wedges

Read:

- Equation (6) measures whether today's fundamental predicts future excess returns.
- Equation (7) measures whether today's fundamental predicts future subjective forecast errors.

- Proposition 1 writes the slope coefficients using true and perceived autocorrelations.
- If the perceived autocorrelation equals the true autocorrelation at all lags, both families of coefficients are zero.

Economic message: Predictability is a diagnostic for autocorrelation misspecification.

5.3 Learning under misspecification and the main characterization

Recall from Section 2 that the true data-generating process can be represented by the collection of parameters $\theta^* = (A^*, B^*, c^*)$, all of which are of dimension n , whereas agents' models are restricted to the set of k -dimensional models, Θ_k . Given an arbitrary model $\theta \in \Theta_k$ for the agents, denote the KL divergence of model θ from the true process by

$$\text{KL}(\theta^* \parallel \theta) = \mathbb{E}^*[-\log f^\theta(x_{t+1} | x_t, \dots)] - \mathbb{E}^*[-\log f^*(x_{t+1} | x_t, \dots)], \quad (10)$$

where f^* is the density of the fundamental under the true data-generating process, f^θ is the agents' subjective density under model θ , and $\mathbb{E}^*[\cdot]$ is the expectation with respect to the true process. It is well known that the KL divergence is always non-negative and obtains its minimum value of zero if and only if densities f^θ and f^* coincide almost everywhere. As such, $\text{KL}(\theta^* \parallel \theta)$ can be interpreted as a measure for the disparity between agents' subjective expectations under model θ and the true process.

The next result characterizes agents' subjective expectations in terms of the KL divergence from the true data-generating process. As discussed in Subsection 2.3, we focus on the outcome of learning under the assumption that agents have access to a long history of the past realizations of the fundamental. This allows us to study the systematic implications of the complexity constraint for agents' subjective expectations, while abstracting from the fluctuations in beliefs that arise due to a finite sample.

Proposition 2. Let $\tilde{\Theta}_k \subseteq \Theta_k$ be an arbitrary compact set with $\bar{\mu}(\tilde{\Theta}_k) = 1$ and let $\tilde{\Theta} \subseteq \tilde{\Theta}_k$ denote a collection of k -factor models with a positive measure under the agents' prior beliefs, i.e. $\bar{\mu}(\tilde{\Theta}) > 0$. If

$$\text{ess inf}_{\theta \in \tilde{\Theta}} \text{KL}(\theta^* \parallel \theta) > \min_{\theta \in \tilde{\Theta}_k} \text{KL}(\theta^* \parallel \theta),$$

then

$$\lim_{t_0 \rightarrow -\infty} \mu_t(\tilde{\Theta}) = 0 \quad \mathbb{P}^* \text{-almost surely.}$$

(a) Proposition 2: Bayesian learning selects the KL-minimizing approximation

Read:

- Proposition 2 shows that posterior mass concentrates on the best feasible k -factor approximation.
- Theorem 1 turns that approximation into perceived autocorrelations and predictability slopes.
- The key inputs are the true ACF and the complexity limit k .
- No extra free parameters are introduced for expectations once k is specified.

Economic message: Bayesian learning does not eliminate misspecification when agents cannot represent the true model.

following result.

Theorem 1. Suppose agents are constrained to k -factor models and let $\{\zeta_t^*\}_{t=-\infty}^\infty$ denote the true autocorrelation of the fundamental.

(1) The autocorrelation of the fundamental according to agents' subjective expectations is

$$\tilde{\zeta}_h = \frac{\sum_{t=0}^\infty (u' \mathbf{M}^{h+1} u)(u' \mathbf{M}^t u)}{\sum_{t=0}^\infty (u' \mathbf{M}^t u)^2} \quad \text{for all } h \geq 0, \quad (11)$$

where $\mathbf{M}$ and u are, respectively, a $k \times k$ stable matrix and a $k \times 1$ unit vector that minimize

$$H(\mathbf{M}, u) = 1 - 2 \sum_{s=1}^\infty \phi_s^{(1)} \zeta_s^* + \sum_{s=1}^\infty \sum_{t=1}^\infty \phi_s^{(1)} \phi_t^{(1)} \zeta_{t-s}^* \quad (12)$$

subject to $\rho(\mathbf{M}(\mathbf{I} - uu')\mathbf{M}') < 1$ and $\phi_s^{(m)} = u' \mathbf{M}^{m-1} [\mathbf{M}(\mathbf{I} - uu')]^{s-1} \mathbf{M} u$.

(2) The coefficients of the return and forecast-error predictability regressions are given by

$$\beta_h^{\text{rx}} = \delta \frac{\zeta_h^* - \sum_{t=1}^\infty \phi_t^{(1)} \zeta_{h-t}^*}{1 - \sum_{t=1}^\infty \delta^t \phi_t^{(1)}} \quad (13)$$

$$\beta_{h,m}^{\text{fe}} = \zeta_{h+m-1}^* - \sum_{t=1}^\infty \phi_t^{(m)} \zeta_{h-t}^*. \quad (14)$$

(b) Theorem 1: model complexity maps into predictability

5.4 Comparative statics and single-factor models

and forecast-error predictability at different horizons, as we do in the remainder of this section.

We start with a simple result that considers the case where agents' models are sufficiently rich to fully capture the statistical properties of the data-generating process.

Proposition 3. *Let k and n denote the number of factors in, respectively, agents' model and the true data-generating process.*

- (1) *If $k \geq n$, then $\beta_h^\alpha = \beta_{h,m}^\alpha = 0$ for all horizons h and all m .*
- (2) *If $k < n$, then there exist h , $\tilde{h}$, and m such that $\beta_h^\alpha \neq 0$ and $\beta_{h,m}^\alpha \neq 0$.*

The first statement establishes that if agents can contemplate models that are as complex as the true model, then there is no return or forecast-error predictability at any horizon. This is a consequence of the fact that agents in our framework are Bayesian with access to a long history of observations. Bayesian updating, together with the assumption that $k \geq n$, implies that agents rule out models that are inconsistent with their past observations in favour of the true data-generating process, while the large sample size guarantees that they do not suffer from finite-sample problems such as overfitting. They recover the underlying data-generating process even if they use models that have too many parameters relative to the true model (i.e. when $k > n$). The second statement of Proposition 3 shows that the converse implication is also true: if there is no predictability, the set of models entertained by agents has to be rich enough to contain the true underlying model. This result follows from the characterization in Proposition 1, which implies that $\beta_h^\alpha = \beta_{h,m}^\alpha = 0$ at all horizons h if and only if the model-implied autocorrelation coincides with the true autocorrelation of the data-generating process at all lags ($\zeta_r = \zeta_r^*$ for all r). But this can happen only if agents are capable of entertaining complex enough models

(a) Propositions 3–4: when predictability vanishes

cally distributed process e_t .¹¹ Note that these coefficients only depend on the statistical properties of the true data-generating process and are independent of agents' subjective model (or its complexity). We have the following result.

Proposition 5. *Let $\tilde{\beta}^{fe} = \sum_{h=1}^{\infty} \phi_h^* \beta_{h,1}^{fe}$, where $\beta_{h,m}^{fe}$ is the coefficient of the forecast-error predictability regression (7). Then,*

- (1) *$\tilde{\beta}^{fe} > 0$ for all $k < n$;*
- (2) *$\tilde{\beta}^{fe}$ is decreasing in k ;*
- (3) *$\tilde{\beta}^{fe} = 0$ for all $k \geq n$.*

This comparative static result captures the intuitive idea that if agents can entertain more complex models with richer time-series dependencies, their forecast errors (averaged over different horizons) become less predictable. More specifically, Proposition 5 establishes that, holding the underlying data-generating process fixed, a particular discounted sum of the term structure of the coefficients of the predictability regression (7) is strictly positive for small values of k , decreases as the agents rely on more complex models, and becomes equal to zero when k exceeds n .

4.3. Single-factor models

While the optimization problem in Theorem 1—which relates return and forecast-error predictability to the primitives of the economy—does not have a closed-form characterization for a general k , it is possible to obtain such a characterization for the case in which agents are restricted to the class of single-factor models. Despite being only a special case, this closed-form characterization is a transparent and easy-to-use result that is informative about how the extent of return and forecast-error predictability varies with horizon h and the statistical properties of the true data-generating process.

Proposition 6. *If agents are constrained to single-factor models, then the coefficients of the return and forecast-error predictability regressions are given by*

$$\beta_h^\alpha = \frac{\delta}{1-\delta} (\zeta_h^\alpha - \zeta_{h-1}^\alpha \zeta_1^*) \quad (15)$$

(b) Propositions 5–6: forecast-error predictability and single-factor formulas

Read:

- If $k \geq n$, rational-expectations predictability restrictions are restored.
- If $k < n$, predictability generally appears.
- Long-run predictability eventually vanishes because the processes are stationary and ergodic.
- Forecast-error predictability declines as k increases.
- For $k = 1$, return and forecast-error slopes have simple closed forms in terms of the true ACF.

Economic message: More sophisticated subjective models are closer to rational expectations. The one-factor case is especially transparent and powerful for the UIP application.

5.5 Identification of k and the UIP mapping

Proposition 7. *Suppose $k = 1$ and $n = 2$. Furthermore, suppose $\mathbf{A}^*$ has two distinct eigenvalues, λ_1 and λ_2 , such that $|\lambda_1| \geq |\lambda_2|$.*

- (1) *If both eigenvalues are real and $\lambda_1 > 0$, then β_h^α never changes sign.*
- (2) *If both eigenvalues are real and $\lambda_1 < 0$, then β_h^α changes sign at every horizon.*
- (3) *If both eigenvalues are complex, then β_h^α changes sign at horizons $h = r\pi/|\varphi|$ for $r \in \mathbb{N}$, where $\varphi = \arg(\lambda_1)$ is the argument of λ_1 in its polar representation.*

That the return predictability pattern depends on the eigenvalues of $\mathbf{A}^*$ is a consequence of the fact that these eigenvalues are what determine the behaviour of the autocorrelation of the data-generating process at different horizons. For example, when $\mathbf{A}^*$ has a complex conjugate pair of eigenvalues, ζ_h^* exhibits an oscillatory behaviour. This pattern is then naturally reflected in the behaviour of β_h^α , as agents who are restricted to one-factor models fit a process with an exponentially decaying autocorrelation to a time series with a more complex autocorrelation.

4.4. Identifying the complexity of agents' model

Our results so far were focused on characterizing the extent of return and forecast-error predictability as a function of the complexity of agents' model. We conclude this section by establishing how one can back out the number of factors in agents' model from their forecasts and the realizations of the fundamental.

Consider the following representation of agents' m -step-ahead forecasts in terms of the history of their observations:

$$\mathbb{E}_t[x_{t+m}] = \sum_{\tau=1}^m \phi_\tau^{(m)} x_{t+1-\tau}, \quad (17)$$

where $(\phi_1^{(m)}, \phi_2^{(m)}, \dots)$ are a collection of coefficients that are independent of the fundamental's realizations. Given these coefficients, let Φ denote the (infinite-dimensional) Hankel matrix with

(a) Propositions 7–8: sign reversals and identifying k

price of the foreign currency. With the discount rate set to $\delta = 1$, equation (2) is then nothing but the (log-linearized) interest rate parity condition:

$$y_t = (i_t^* - i_t) + \mathbb{E}_t[y_{t+1}]. \quad (19)$$

Note that, as in Subsection 2.1, the expectation in (19) denotes investors' subjective expectations, which may differ from those arising from the true data-generating process. Finally, equation (3) is simply the definition of currency excess returns:

$$rx_{t+1} = y_{t+1} - y_t + (i_t^* - i_t).$$

We reproduce the empirical findings on UIP violations at different horizons by following Engel (2016) and building a trade-weighted average exchange rate and interest rate differential relative to the US for the following countries: Australia, Canada, the Eurozone (Germany before its introduction), New Zealand, Japan, Switzerland, and the United Kingdom. The weights are constructed as the value of each country's exports and imports as a fraction of the average value of trade over the seven countries. Monthly exchange rate data is from Datastream and interest rate differentials are calculated using covered interest rate parity (CIP) from forward rates, $i_t^* - i_t = f_t - y_t$, also available from Datastream. We then run the following family of regressions:

$$rx_{t+h} = \alpha_h^\alpha + \beta_h^\alpha (i_t^* - i_t) + e_{t,h},$$

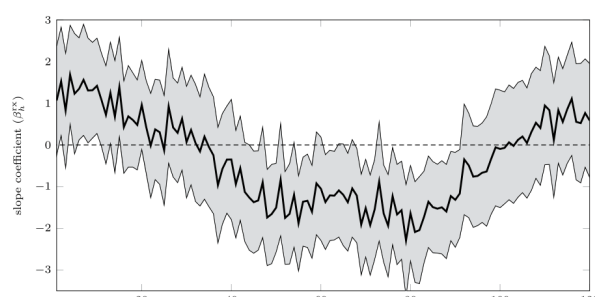
(b) UIP mapping and horizon regression

Read:

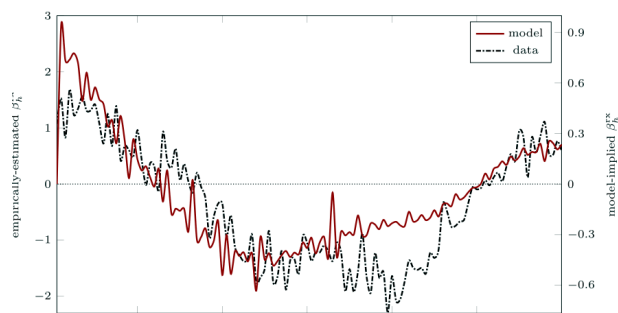
- Proposition 7 links sign reversals to the eigenvalue structure of the true process.
- Proposition 8 shows that the rank of a Hankel matrix formed from forecast coefficients identifies k .
- In UIP, x_t is the interest-rate differential and y_t is the log exchange rate.
- The return regression becomes a horizon regression of currency excess returns on the current interest-rate differential.

Economic message: The same abstract model becomes a concrete test of currency return predictability and forecast errors.

5.6 Figure 1 + Figure 2: UIP violation and one-factor model fit



(a) Figure 1: empirical UIP violation at different horizons



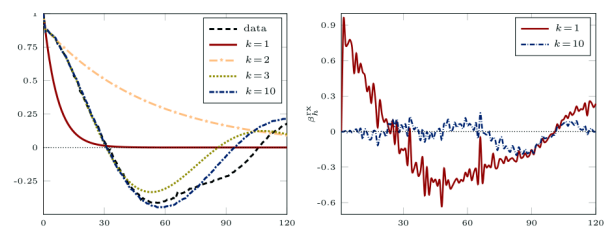
(b) Figure 2: empirical vs one-factor model-implied return predictability

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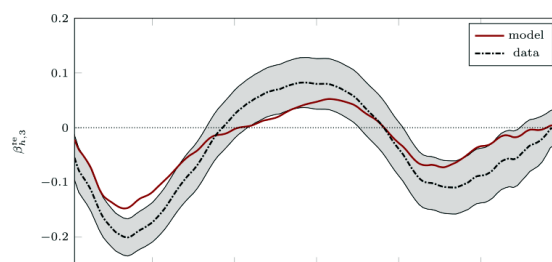
- Figure 1 shows that high interest-rate currencies earn positive excess returns at short horizons.
- The coefficient turns negative around medium-long horizons, capturing the predictability reversal puzzle.
- At very long horizons, the coefficient moves back toward zero.
- Figure 2 overlays the empirical term structure with the one-factor model-implied term structure.
- The one-factor model captures the short-run positive coefficient and the long-run sign reversal.

Economic message: A constrained one-factor forecasting model can jointly explain the forward discount puzzle and predictability reversal.

5.7 Figure 3 + Figure 4: complexity and survey forecast errors



(a) Figure 3: ACF and model-implied return predictability for different k



(b) Figure 4: survey forecast-error predictability

Read:

- In Figure 3, a larger k can fit the empirical autocorrelation function better.
- However, a high-dimensional model also produces return-predictability coefficients closer to zero.
- The one-factor model does not match the ACF perfectly, but it generates the observed return-predictability pattern.
- Figure 4 shows that survey-based forecast errors are also predictable in a pattern close to the one-factor model.

Economic message: Fitting the ACF too well eliminates the expectation wedge. The observed return and forecast-error patterns point to a low-dimensional subjective model.

5.8 Appendix Figure A.1: heterogeneous agents

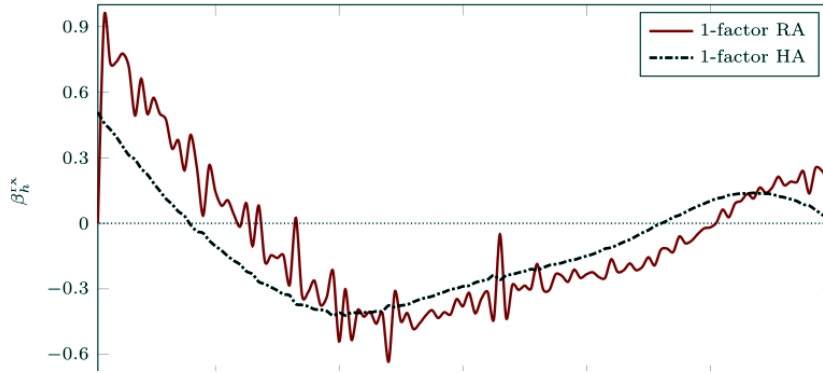


Figure 9: Figure A.1: model-implied return predictability with 90% rational agents and 10% behavioural agents.

Read:

- The appendix considers an economy with both rational and behavioural agents.
- Even when only 10% of agents use the single-factor constrained model, model-implied predictability remains visible.
- The representative-agent behavioural benchmark and the heterogeneous-agent case show similar qualitative patterns.

Economic message: Small amounts of belief misspecification can matter for equilibrium returns when expectations enter prices.

6 Short summary

- The paper studies asset pricing when agents are Bayesian but constrained to low-complexity statistical models.
- The true process has n hidden factors; agents can use at most k factors.
- If $k \geq n$, the rational-expectations benchmark is recovered.
- If $k < n$, agents learn the KL-minimizing low-dimensional approximation to the true process.
- Prices depend on subjective expectations, so the approximation error appears in excess returns.
- Return predictability and forecast-error predictability are both characterized by the gap between true and perceived autocorrelations.

- The main theorem maps the empirical autocorrelation function and the model dimension k into the full term structure of predictability coefficients.
- Single-factor models produce closed-form coefficients and can generate sign reversals when the true ACF is more complex.
- In the UIP application, a one-factor model explains both short-run positive currency excess returns and long-run predictability reversal.
- Survey forecast errors of interest-rate differentials show a pattern consistent with the one-factor model.
- A rank test based on survey forecasts cannot reject a one-factor subjective model.
- The paper’s core lesson is that simple models can endogenously generate systematic belief errors and return predictability without assuming non-Bayesian updating.

7 Conclusion

7.1 Core conclusion

Molavi, Tahbaz-Salehi, and Vedolin show that limits on model complexity can be a disciplined source of asset-pricing anomalies. Agents are rational conditional on what they can represent, but when their subjective model class is too small, they learn the wrong low-dimensional approximation. Prices then reflect systematically misspecified expectations.

7.2 Economic intuition

The economic intuition is that agents compress a complex time series into too few state variables. This compression distorts perceived autocorrelation. If the true process has richer medium-horizon dynamics than the subjective model can represent, agents over- or under-predict future fundamentals at different horizons. Since returns are forecast revisions, these predictable forecast errors become predictable returns.

7.3 Takeaway

The paper provides a unified expectation-based explanation for UIP violations: high interest-rate currencies can earn positive short-run returns and negative longer-horizon returns because investors’ low-dimensional models misread the intertemporal dynamics of interest-rate differentials. More broadly, the paper turns model complexity into an asset-pricing state variable.